

Mahindra University Hyderabad

Ecole Centrale School of Engineering

Minor-II Examination

Program: B. Tech. Branch: CAM Year: II Semester: I

Subject: Design and Analysis of Algorithms (MA 2107)

Date: 31/10/2025

Time Duration: 1.5 Hours

Start Time: 2:00 PM

Max. Marks: 20

Instructions:

1. There are 4 questions, all of which are compulsory.
2. Justify your response whenever necessary. Guesswork will not be considered in evaluation.
3. Answers without proper numbering will not be evaluated.

1. Answer the following questions:

5 M

a) Define spanning tree for a graph G .

2M

b) Write Prim's Algorithm.

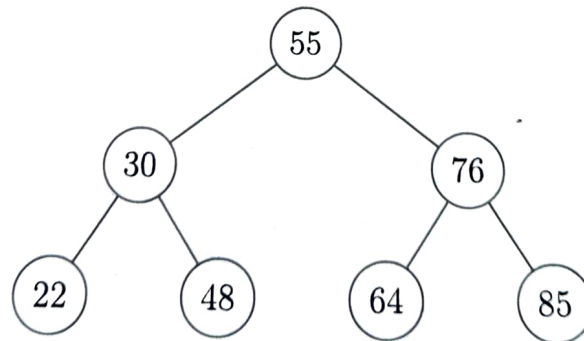
3M

2. a) What is the main difference between a Binary Tree and a Binary search tree? 2M

5 M

b) Consider the following binary tree. Is it a Binary Search Tree?

1M



c) Consider the above Tree [in 2b], what is the pre-order traversal of it?

2M

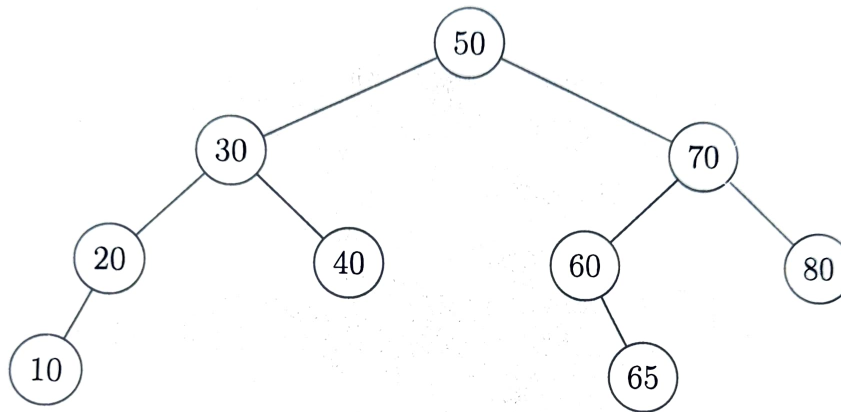
3. a) Define AVL Tree.

2M

5 M

b) Consider the following AVL tree. Delete 40 from it.

3M



4. Insert 15, 22, 30 into an empty Red Black tree. Show all the steps.

5 M
